TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Neidhardt, Joerg	**Email**	Joerg.neidhardt@fep.fraunhofer.de
Institute Name	Fraunhofer Institute for Electron Beam and Plasma Technologies			
Address	Winterbergstr. 28, 01277 Dresden, Germany			

2. Technology Overview

Title of Proposed Technology	Large Area Processing of Ultra-Thin Flexible Glass			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (x), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials		Electrical, Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Surface Treatment	*Coating / Method Technologies*	*Display*	*Display Manufacturing Equipment*
Concept of Proposed Technology	*FEP provides surface engineering and coating platforms for pilot scale **in-line sheet-to-sheet** as well as **roll-to-roll** coating and treatment of thin and **ultra-thin flexible glass** (UTG) down to 30 µm thickness. Optical layer stacks for **smart windows, displays** as well as **space and aviation** applications can be designed, screened and provided on sheets of up to 600x1200 mm² and/or rolls up to 300 mm width. Demonstrated use cases:* - *Transparent conductive oxides for photovoltaic and smart windows* - *Active thermochromic layers for smart windows* - *Broadband antireflection for lightweight PV for aviation and space* - *Coatings for wavelength-selective reflection, e.g., for head-up displays* - *Self-cleaning, anti-smudge and anti-scratch functionalization by UTG on consumer surfaces* - *Strengthening of UTG by inline flash lamp annealing*			

(max 10 pages)

Rationale for Joint R&D Project	*Fraunhofer FEP provides the process know how and capabilities for a broad range of applications up to a pilot scale. The application and the system know how is typically provided by the partner in a collaborative approach.*
Key Words	*1. Ultra-thin flexible glass* *2. in-line magnetron sputtering* *3. smart windows* *4. flexible displays* *5. photovoltaic for space and aviation*

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (7)
R&D Objectives	*Layer stack design, development and demonstration* *Scaling of technology and proof of principle* *Validation of pilot production and specification of production technology*			
R&D Target Specifications	*To be defined by Korean project partner*			
R&D Plan	*Strongly dependent on the final application and to be agreed upon together with Korean project partner*			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Research and Development infrastructure (coating equipment, personnel) Coating and substrate analysis Scientific support for coating design and implementation
Expected R&R of Korean SME Partner	Target application and requirements for substrate and coating properties Integration and validation of coated UTG substrates Joined commercial analysis (cost of ownership depending on required throughput)

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Novel products based on ultrathin flexible glass as a functionalized substrate

6. References

(1) W. Langgemach und E. Rädlein. A new method for evaluating the influence of coatings on the strength and fatigue behavior of flexible glass. In: J. Electron. Mater. 7 (2024), S. 17168. doi: 10.1007/s11664-024-11015-x.

(2) W. Langgemach, A. Baumann, M. Ehrhardt, T. Preußner und E. Rädlein. The strength of uncoated and coated ultra-thin flexible glass under cyclic load. In: AIMS Mater. Sci. 11 (2024), S. 343–368. doi: 10.3934/matersci.2024019.

(3) W. Langgemach, M. Fahland und E. Rädlein. The influence of sputter-deposited indium tin oxide thin films on the strength of ultra-thin glass. In: Open Ceram. 21 (2025), S. 100710. doi: 10.1016/j.oceram.2024.100710.

(4) W. Langgemach. Investigations of the influence of sputter-deposited thin films on the strength and the fatigue behavior of ultra-thin glass. Bd. 28. Werkstofftechnik aktuell. Ilmenau: Universitätsverlag Ilmenau, 2025. doi: 10.22032/dbt.67102.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)